The Banach–Tarski Paradox

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1 Introduction

The Banach–Tarski paradox is a famous dilemma in modern mathematics. It asserts that a solid sphere or any other object that exists in a three-dimensional space can theoretically be divided so that these pieces after being moved around by rotations and translations alone, can be reassembled into *two* spheres identical to the original one.

What makes this difficult to believe is that there is no distortion involved. This seemingly violates not only geometry, but common sense itself. How can an object become two objects of equal size without creating any additional material?



Figure 1: A schematic illustration of the Banach–Tarski paradox. A solid sphere is decomposed into a finite amount of disjoint pieces. Using only rotations and translations it can be reassembled into two spheres congruent to the original.

1.1 Volume and Additivity

1.1.1 Volume Preservation

We consider volume to be the 'space' that an object occupies in the third dimension. It is comparatively similar to 'area' in two-dimensional spaces. An object's volume does not necessarily need to be conserved, evident by the properties of densities, where the same mass can be compacted in different volume amounts.

However, in real world situations, volume behaves predictably. For example, if a log of wood is cut into two pieces, its volume will not change. This is because for volume to change in this regard, it requires distortion through dilation or reduction. If we were to consider an object in a two dimensional space, such as the surface of a pizza, we can cut it as many times as we would like, but we would still have the same total area if we were to reassemble the pieces.

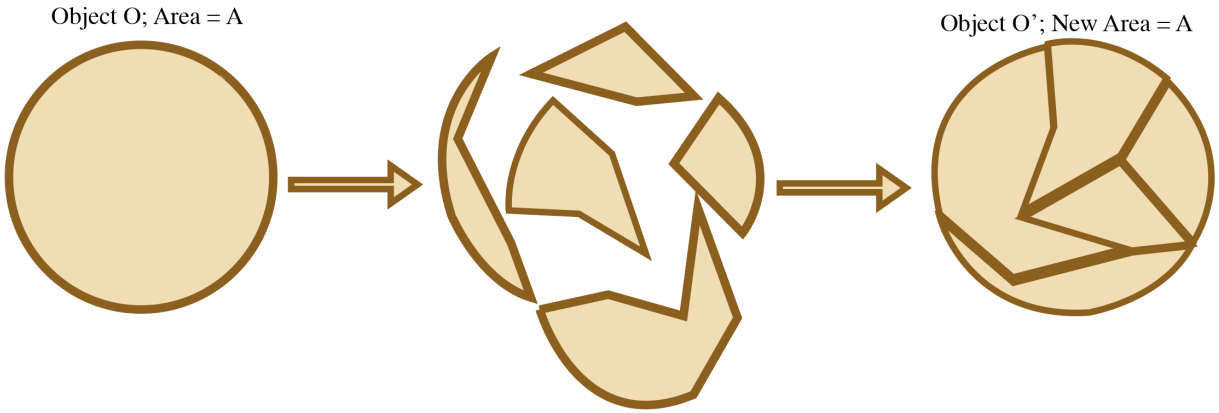


Figure 2: Cutting and reassembling a measurable object (like this 2D shape) preserves its total area. Our physical intuition about volume is rooted in the behavior of such well-behaved sets

We can distort the size of one of these pieces, but then the area of the pizza's surface would no longer be conserved. It does not seem feasible to have two objects with same density *and* volume simply through rotations and translations of pieces from one object.

1.1.2 Defining Volume

We must clarify what exactly we define as 'volume'. The volumes of shapes from Euclidean geometry (balls, cubes, polyhedra, etc.) satisfy the property of *finite additivity*: if two objects are not overlapping, the volume of their union is the sum of their individual volumes.

However, this property assumes *every* subset of three-dimensional space has a well-defined volume. This assumption can be made in regular geometry because typically the only sets we encounter are orderly ones that have regions with clean boundaries. But once we consider arbitrary sets, we can no longer guarantee this assumption. Some sets are so irregular that no consistent notion of volume can be assigned to them at all; we define these sets as *nonmeasurable*.

1.2 Overview

Therefore, the Banach–Tarski paradox does not necessarily have to be a direct contradiction of physics' principle of mass conservation. Rather, it contradicts our intuition that every conceivable set must behave like a physical object. If we assume the Axiom of Choice, then it is possible for three-dimensional space to contain subsets that defy regular geometry.

In the following section, we discuss in more detail the distinction between measurable and nonmeasurable sets. In Section 3 we introduce the Axiom of Choice and its significance in creating non-measurable sets. We will also look at the Vitali Set as a simpler, one-dimensional analogue of the paradox. Section 4 will briefly present the Banach–Tarski theorem itself, and Section 5 will go over concluding remarks and discussion.

2 Measurable and Non-Measurable Sets

2.1 Definition of Measure

When we discuss the notion of *measure*, we mean it as a way to assign a consistent notion "size" to sets. This can come in the form of length, area, or volume.

A measure can be thought of as a function that assigns a nonnegative number to each set in order to represent its "size". For example, on the real number line, the interval $[0, 1]$ has length 1, while $[0, 2]$ has length 2. This is an example of 'measure' for real numbers. However, we would like our definition of measure to behave reasonably for all subsets of $\mathbb{R}$ (as well as higher-dimensional spaces).

For our definition of measure to be useful, a measure should fulfill these particular properties:

- **Non-negativity:** The measure of any set is nonnegative.
- **Countable additivity:** If a set can be expressed as a countable union of disjoint subsets, its measure is equal to the sum of their measures.
- **Translation Invariance** Shifting a set does not change its measure. That is, if A is measurable and t is any real number, then $\mu(A + t) = \mu(A)$. In some cases, this is not considered necessary for a measure, although we will use it for our purposes. The notion of translation may be different for different spaces, such as 1-dimensional or 2-dimensional space.

These properties reflect our physical intuition; cutting or rearranging doesn't change an object's total volume. This is significant because the consistency with physical logic makes the notion of measure meaningful even in higher-dimensional spaces which are difficult to imagine.

2.2 The Lebesgue Measure

The Lebesgue measure was developed in order to handle the measure of more complicated sets. It is mainly based in approximation: for any set $A \subseteq \mathbb{R}$, we can try to cover the set with a collection of intervals whose total length will get as close as possible to the "size" of A.

The smallest total length of a collection of coverings is called the *outer measure* of A, which we define as $\mu^*(A)$. $\mu^*(A)$ represents the least amount of total length needed to completely cover A with disjoint @intervals, regardless of how irregular the set is.

However, not every set can be assigned a measure consistently using only outer measure. Some sets interact with other subsets in ways that make their size ambiguous. Henri Lebesgue addressed this by introducing *Carathéodory's condition* which determines which sets can be considered measurable. A set A is said to be *Lebesgue measurable* if, for every set $E \subseteq \mathbb{R}$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).$$

This condition ensures that A divides any other set E without creating contradictions in how we compute total measure. It basically ensures that if we cut any set by A, we are not interfering with the measurements of the other sets.

The Lebesgue measure, which we denote as μ, is then the outer measure which is restricted to all measurable sets. For intervals, it is aligned with our typical notion of length.

$$\mu([a, b]) = b - a.$$

This definition also extends into more complex sets, such as countable unions of intervals and most regions that one could encounter in analysis and geometry.

The Lebesgue Measure's generality allows us to talk about size across a large range of sets while still preserving the properties that define measurable sets. However, there are some subsets of $\mathbb{R}$ in which no measure satisfying these properties can consistently assign a size. These are also known as *nonmeasurable sets.*

2.3 Conceptual Limits of Measure

The Lebesgue measure seems to be an appropriate and flexible method to assign size to subsets of $\mathbb{R}$. Therefore, it is natural to wonder if the Lebesgue measure can be applied to *every* subset of $\mathbb{R}$.

However, it is not possible to do this. This is not a result of the Lebesgue measure failing. Rather, it is because no function can consistently assign a notion of size to all subsets of $\mathbb{R}$ while also preserving the properties of measure.

Consider the three properties we have previously defined: non-negativity, countable additivity, and translation invariance. Individually, these properties are reasonable and together they are the basis for our natural geometric/physical understanding.

However, these properties cannot all be true for every subset of $\mathbb{R}$. If we try to make *every* subset of $\mathbb{R}$ measurable, either translation invariance or additivity must fail. Conversely, if we try to preserve translation invariance and countable additivity, then some subsets of $\mathbb{R}$ must be non-measurable.

This contradiction occurs because $\mathbb{R}$ contains a large amount of internal symmetry. Translations can shift sets in infinitely many ways. When these shifted sets interact with other sets, additivity imposes rigid constraints on their sizes. Certain subsets can be arranged through translations alone so that they overlap in highly nontrivial ways, creating contradictions if every set were assigned a measure.

Essentially, non-measurable sets exist so that measurable sets, which are defined by their size, additivity, and being translation-additivity, can inform our notion of size consistently. Lebesgue measure does not fail because it excludes some sets, instead it succeeds *because* it does do that.

2.4 Nonmeasurable Sets

Unlike measurable sets, nonmeasurable sets can't be characterized with boundaries or shapes. Rather, they are defined by how their points are *distributed* across space. Nonmeasurable sets are typically constructed by selecting points from collections of equivalence classes (groups of points considered "equivalent" under some rule), rather than by carving out regions of space geometrically.

As a result, nonmeasurable sets are very fragmented. Because they are scattered throughout $\mathbb{R}$, there is no interval of values that can meaningfully capture their size.

Additionally, nonmeasurable sets are very sensitive to translations. Shifting a nonmeasurable set can produce translated copies that interact in a way that conflicts with additivity. If we attempt to assign a consistent measure to the set, translation invariance would require all copies to have the same measure. However, then countable additivity would demand that certain unions of these copies have a measure that can't correspond to the actual amount of space they cover.

Essentially, nonmeasurable sets can't be defined by any explicit rule or formula. Rather, they come from unrestricted selection processes that choose elements from infinitely many overlapping collections without regard to spatial structure. In a sense, nonmeasurable sets are like "weird selections".

To form such sets, one must be able to make infinitely many independent choices without a specific rule. The principle that allows us to do this is known as the *Axiom of Choice*; which is very significant to the existence of nonmeasurable sets.

3 The Axiom of Choice and the Vitali Set

3.1 The Axiom of Choice

The Axiom of Choice (AC) is a principle in set theory that allows us to select elements from arbitrary collections of sets, even if there is no explicit rule given for the selection. This can formally be stated as:

$$\forall \mathcal{C}\,(\emptyset \notin \mathcal{C} \implies \exists f : \mathcal{C} \to \bigcup \mathcal{C} \text{ such that } f(X) \in X \ \forall X \in \mathcal{C}).$$

- $\mathcal{C}$ is defined as any collection of non-empty sets.
- f is defined as a *choice function* that selects one element from each set $X \in \mathcal{C}$.

When $\mathcal{C}$ is finite, a choice function can be constructed explicitly. However, for infinite collections, particularly uncountable ones, AC ensures the existence of a choice function even when there isn't any explicit rule for making the selections.

This principle is important for constructing certain sets that cannot be defined explicitly by a formula, such as the Vitali set. To construct a Vitali set, it is required to make infinitely many independent choices, one from each equivalence class of real numbers modulo $\mathbb{Q}$. The Axiom of Choice is what allows us to make such a selection.

3.2 Equivalence Classes Modulo Rationals

To construct the Vitali set, we first define an equivalence relation on $\mathbb{R}$. For $x, y \in \mathbb{R}$, we define this as:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This is an equivalence relation because is satisfies the following properties:

- Reflexive: $x - x = 0 \in \mathbb{Q} \implies x \sim x$.
- Symmetric: If $x \sim y$, then $x - y \in \mathbb{Q} \implies y - x = -(x - y) \in \mathbb{Q} \implies y \sim x$.
- Transitive: If $x \sim y$ and $y \sim z$, then $x - z = (x - y) + (y - z) \in \mathbb{Q} \implies x \sim z$.

The set of equivalence classes under this relation is denoted as $\mathbb{R}/\sim$. Each equivalence class is a dense subset of $\mathbb{R}$ consisting of all real numbers differing by a rational number.

Each equivalence class can be thought of as a "thread" running through the real line. No matter how far you move along the line, you will always find points from each class that are spaced by rational distances. For the Vitali Set, we will select exactly one representative from each class.

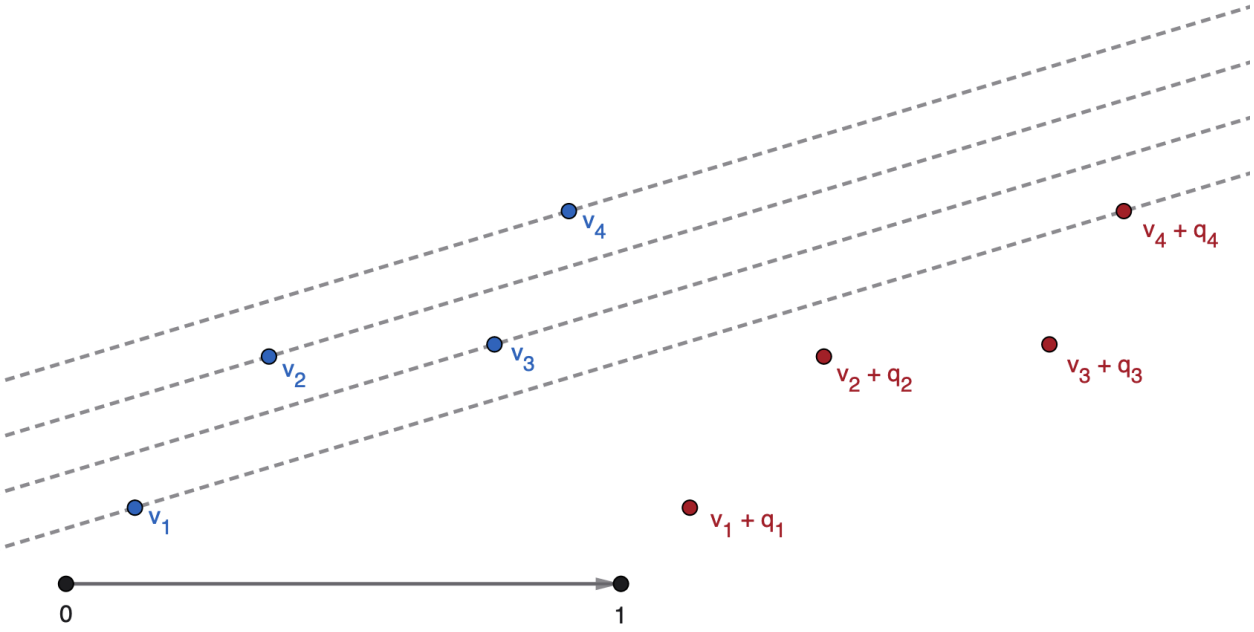


Figure 3: A schematic illustration of equivalence classes of $\mathbb{R}$ modulo $\mathbb{Q}$. Each slanted line represents one equivalence class. Selecting a single point from each class produces the Vitali set V. Horizontal translations of V by rational offsets generate disjoint shifted copies whose union covers an interval, visualizing the source of non-measurability.

3.3 Construction of the Vitali Set

The Vitali set $V \subset [0, 1]$ is constructed by selecting one representation from each equivalence class in $\mathbb{R}/\sim$ that intersects $[0, 1]$. We define this formally using the Axiom of Choice as:

$$V = \{v_\alpha \in \alpha : \alpha \in (\mathbb{R}/\sim) \cap [0, 1]\}.$$

There isn't an explicit formula for the elements of V, so the selection must rely entirely on the Axiom of Choice. This means that the Vitali set can't be constructed explicitly ; it is inherently non-measurable.

3.4 Non-Measurability of the Vitali Set

To see why V is non-measurable, let us consider the set of all rational shifts of V within $[0, 1]$:

$$V_r = V + r \quad \text{for} \quad r \in \mathbb{Q} \cap [0, 1).$$

These shifted sets are disjoint. If $x \in V_r \cap V_s$ for $r \neq s$, then $x - r, x - s \in V$ and $x - r \sim x - s \implies r - s \in \mathbb{Q}$. This contradicts the construction of V, which includes exactly one representative from each equivalence class.

Furthermore, the union of all such shifts essentially covers $[0, 1]$:

$$[0, 1] \subseteq \bigcup_{r \in \mathbb{Q} \cap [0,1)} V_r \subseteq [0, 2].$$

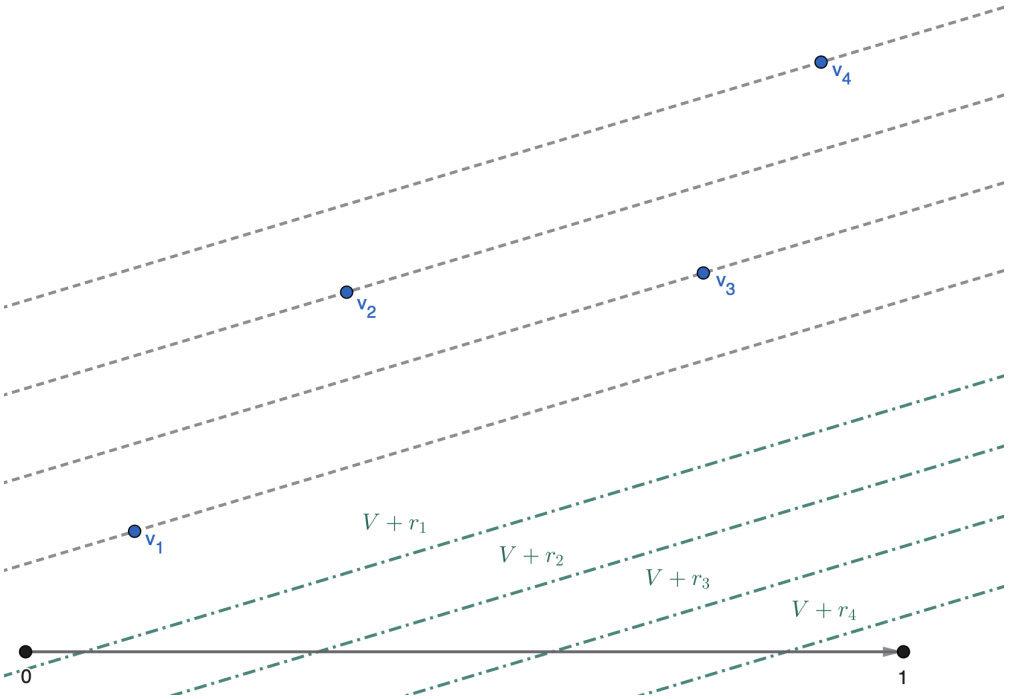


Figure 4: Disjoint rational shifts of the Vitali set. Each copy is non-overlapping, but their union covers $[0, 1]$ which shows how it is not feasible to assign a consistent measure.

Hypothetically, let us assume that V is Lebesgue measurable with measure $\mu(V)$. By translation invariance all shifted sets have the same measure: $\mu(V_r) = \mu(V)$. By countable additivity, we have:

$$\mu\Big(\bigcup_{r \in \mathbb{Q} \cap [0,1)} V_r \Big) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(V_r) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(V).$$

Since $\mathbb{Q} \cap [0, 1)$ is countably infinite, the sum is 0 if $\mu(V) = 0$, or it diverges to infinity if $\mu(V) > 0$. Both possibilities contradict the fact that the union is contained in the finite interval $[0, 2]$. Therefore $\mu(V)$ can't exist, and V is non-measurable.

4 The Banach–Tarski Theorem

4.1 Statement of the Theorem

Formally, the Banach–Tarski theorem states:

Let B be a solid ball in $\mathbb{R}^3$. There exists a decomposition of B into finitely many disjoint sets $A_1, \ldots, A_n$ such that, using only rotations and translations, these pieces can be reassembled to form two identical copies of B.

Equivalently, there exist rigid motions $T_1, \ldots, T_n$ such that

$$\bigcup_{i=1}^{n} T_i(A_i) = B \cup B',$$

where B' is congruent to B and $A_i \cap A_j = \emptyset$ for $i \neq j$.

We have mentioned previously how this defies physical logic in regards to volume preservation. However, this becomes mathematically possible due to properties of the sets A_i. They are non-measurable, can be infinitely fragmented, and exist due to the Axiom of Choice.

4.2 Group Actions and Orbits

The key to the paradox lies in group theory. Consider the group of rotations in $\mathbb{R}^3$, which we denote as $SO(3)$. An important fact about this group, is that it contains a *free subgroup* F_2 generated by two rotations r_1 and r_2:

$$F_2 = \langle r_1, r_2 \rangle = \{r_1^{n_1} r_2^{m_1} \cdots r_1^{n_k} r_2^{m_k} : n_i, m_i \in \mathbb{Z}\},$$

where no non-trivial combination of these rotations equals the identity.

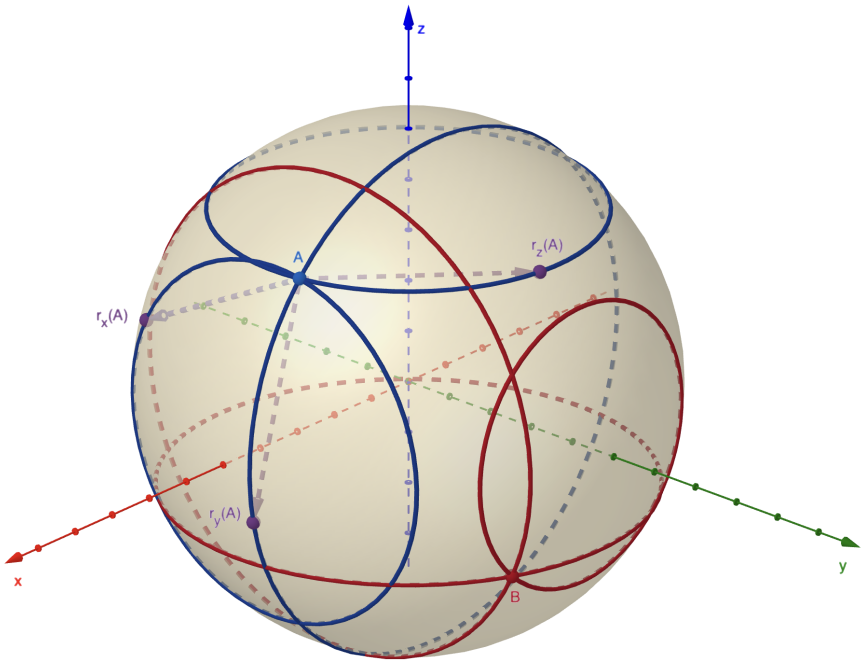


Figure 5: Illustration of orbits of points under rotations. Point A is moved along three rotational axes (x, y, z) to form its orbit. Point B behaves similarly, moving along its own orbit. These orbits are what allows us to select representative points in the Banach–Tarski construction.

Think of each point in the ball as being "moved around" by combinations of the basic rotations. The set of all positions a point can reach with these movements is called its orbit. Almost all points in the ball can be grouped into distinct orbits, with no overlaps between different orbits (with exceptions, like the center).

By picking one point from each orbit, we create a set of representatives that will serve for reconstructing the ball.

For any point $x \in B$, its *orbit* under F_2 is defined as

$$\mathcal{O}(x) = \{g \cdot x : g \in F_2\}.$$

These orbits partition (almost all of) the ball into disjoint equivalence classes under the action of F_2. Using the Axiom of Choice, we select a single representative from each orbit to form a set C. Then the ball can be expressed as

$$B = \bigcup_{g \in F_2} g \cdot C,$$

where the union is disjoint except possibly for a set of measure zero (e.g., the center of the ball).

4.3 Decomposition of the Construction

We can divide the construction into several steps:

1. **Select a free subgroup of rotations.** Let $F_2 = \langle r_1, r_2 \rangle$, generated by two carefully chosen rotations. This subgroup is "paradoxical," meaning that it allows a decomposition similar to the Vitali set construction in 1D.

2. **Partition the ball into orbits.** For each point $x \in B$ outside a negligible set (e.g., the center), its orbit under F_2 defines an equivalence class:

   $$x \sim y \iff \exists g \in F_2 \text{ such that } g \cdot x = y.$$

   The collection of these orbits partitions the ball into infinitely many disjoint classes.

3. **Apply the Axiom of Choice.** Using AC, we pick a single representative from each orbit. This gives us a set C of "choice points."

4. **Form the paradoxical pieces.** By applying group elements $g \in F_2$ to C in certain combinations, we generate disjoint subsets $A_1, \ldots, A_n$ whose union is B and whose rotated/translated copies can form two balls:

   $$B = \bigcup_{i=1}^{n} A_i, \quad B \cup B = \bigcup_{i=1}^{n} T_i(A_i).$$

4.4 Measure-Theoretic Implications

Each piece A_i is non-measurable. If we attempted to assign a Lebesgue measure $\mu(A_i)$, translation invariance and countable additivity would force a contradiction:

- Let $\mu(B) = v$. By translation invariance, each rotated copy $T_i(A_i)$ would have measure $\mu(A_i)$.
- Countable additivity implies the total measure of the union of copies should equal the sum of their measures:

  $$\sum_{i=1}^{n} \mu(A_i) \stackrel{?}{=} \mu\Big(\bigcup_{i=1}^{n} T_i(A_i) \Big) = \mu(B \cup B') = 2v.$$

- But the union of the pieces also equals $B \cup B$, which has a measure of $2v$. This is a contradiction, as the sum $\sum_{i=1}^{n} \mu(A_i)$ cannot equal v and $2v$ under the rules of Lebesgue measure.

So, the paradox does not violate mathematical logic. It can be fully justified because the sets involved here are non-measurable. Measure theory here "fails", but it is in a controlled, understood manner.

5 Conclusions

5.1 Summary

The Banach–Tarski paradox challenges the idea that volume is an intrinsic property of all subsets of space. This intuition for volume is grounded in our real setting in which volume behaves predictably under cutting, rearrangement, and rigid motion. However, once arbitrary subsets are allowed, this assumption no longer holds true.

Using measure theory, we saw that the Lebesgue measure provides a consistent notion of size only by excluding certain pathological sets. The existence of non-measurable sets is not accidental, but rather a consequence of preserving translation invariance and countable additivity. The Axiom of Choice enables the construction of such sets by allowing unrestricted selection from infinitely many equivalence classes, as illustrated by the Vitali set.

In three dimensions, the same ideas manifest through group actions of rotations. The Banach–Tarski Paradox's decomposition relies on the symmetry of $\mathbb{R}^3$, where free subgroups of $SO(3)$ allow space to be partitioned into non-measurable pieces that defy traditional ideas on volume.

5.2 Discussion

Naturally, the paradox is not meant to literally imply physical objects can duplicate, nor does it invalidate conservation of volume as a useful concept. Rather, it shows the limitations in how we apply our definitions. Volume behaves well precisely because it is not defined everywhere for all possible subsets.

Through the Banach–Tarski paradox, it can be seen that by accepting the Axiom of Choice we can gain tools at the cost of admitting objects that resist intuition and visualization. The paradox is both a result that is logically unavoidable and one that is conceptually disturbing to our intuition.

In this way, the paradox serves less as a contradiction and more as a clarification. It reminds us that mathematics is not necessarily constrained to mirror the rules of the physical world, and that the price of generality can often be strangeness.

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